Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building object-oriented applications, simulation pipelines, and test-automated systems, with strong grounding in **C++**, algorithms, and continuous improvement of complex software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Designed and optimized multi-tenant relational schemas across **Azure SQL, PostgreSQL, and SQL Server** for **500+ users** and **10K+ structured records**, reducing dashboard latency by **71%** (**1.2s to 350ms**).
- Built robust backend workflows with **FastAPI**, asynchronous job execution, and cloud storage integrations, processing **5K+ monthly jobs** with **99%+** reliability and structured failure isolation.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Built simulation and model-training pipelines for autonomous driving using a **Decision Transformer**, processing **100K+ trajectory logs** and improving long-horizon planning quality by **18%** in closed-loop evaluation.
- Architected a spatial-temporal encoder combining HD-map geometry and trajectory context through multi-head attention, improving closed-loop planning stability by **15%** and reducing off-road behavior in simulation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform application software with reusable OOP modules and **20+** automated unit/integration test suites using **Jenkins, PyTest, and Mockito**, validating releases across Windows, Linux, and macOS.
- Optimized APIs and ETL pipelines with **AWS Glue, S3, and design patterns**, improving response times by **30%**, increasing pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **microservice-based application platform** with **MERN**, WebSockets, and Azure SQL, synchronizing **2M+ records** across locations and supporting **8** operational workflows for allocation, billing, and notifications.
- Automated build, test, and deployment workflows with **Jenkins, Docker, Jest, Mocha, and Selenium**, improving release validation to **95%+** and reducing resource wastage by **42%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built low-latency **Flask** APIs and real-time Firebase workflows for a cross-platform service platform handling **50K+ records**, improving booking confirmation speed by **40%** with sub-**200ms** API performance.
- Designed payment, coupon, and analytics abstractions with **Databricks, Spark, and Spring Boot-based services**, reducing manual refund handling by **70%** and improving operational visibility.

Skills

Languages: C++, Java, Python, JavaScript/TypeScript, SQL
Software Engineering: Object-Oriented Design, Algorithms, Data Structures, Design Reviews, Problem Solving, Agile/Scrum
Systems & Tools: Simulation Pipelines, FastAPI, Flask, Spring Boot, Jenkins, Docker, Git, CI/CD
Data & Platforms: PostgreSQL, MySQL, SQL Server, Azure SQL, AWS, Databricks, Spark, Firebase

Projects

CPU-scheduling Simulator Live | *C++, React.js, EJ-2*
- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** scheduling algorithms with interactive visualization, reducing debugging time by **20%** and improving average turnaround time by up to **25%** across **1K+** simulations.

Chat2SQL GitHub | *Java, TypeScript, Docker, PostgreSQL*
- Built a SQL development workspace supporting **10+** databases, enabling schema exploration, structured query execution, and cross-service development across Java and TypeScript modules.

Big-Brain Live | *PostgreSQL, FAISS, Pinecone, REST APIs*
- Built a document retrieval platform over **50K+ records** with sub-second search, metadata persistence, and incremental indexing, reducing retrieval latency by **40%** for analysis-heavy user workflows.